

Fixed Cutoffs of the Summing-Basis Weak-Compactness Index

Upgraded conditional packet for arXiv:1507.06285

June 22, 2026

Status

Conditional reduction, not a full solution. This packet upgrades the earlier James-to-wide-(s) bridge packet for Question 9.3 of Beanland–Causey–Freeman–Wallis [1]. The finite levels are already settled by the companion finite-rank packet. The source paper proves that the cutoff $\text{WC} \leq \omega$ is the super weakly compact ideal and that $\text{WC} < \omega_1$ is an ideal. The present packet records a sharper conditional route for higher fixed cutoffs.

The remaining dependency is the explicit tree-Ramsey *summable-fusion* property $\text{SF}(\lambda)$. If $\text{SF}(\lambda)$ holds, then

$$\mathcal{W}_\lambda = \{A : \text{WC}(A) \leq \lambda\}$$

is an operator ideal. The literature supplies the corresponding fixed-mesh pair stabilization at the natural candidate ordinals $\lambda = \omega^{\omega^\eta}$, but fixed mesh alone is not enough.

The supplied full report is preserved in the packet’s **evidence/** directory. The previous conditional packet is preserved in the packet’s **history/** directory.

1 Source Question

The source paper is Beanland–Causey–Freeman–Wallis, *Classes of operators determined by ordinal indices*, arXiv:1507.06285 [1]. It defines the weak-compactness index $\text{WC}(A, X, Y)$ from trees whose images dominate the summing basis of c_0 , then asks:

Question 9.3 (Beanland–Causey–Freeman–Wallis). For which ordinals ξ is the class of operators

$$A : X \rightarrow Y, \quad \text{WC}(A, X, Y) \leq \xi,$$

an ideal?

model indices. We have seen that a better estimate is possible for $\mathbf{NP}_\infty$, and for the $\mathbf{NP}_1$ index when the spaces involved have unconditional bases.

9.2. Weak compactness. Let (s_i) be the summing basis for c_0 . It follows from standard techniques modifying James's characterization of reflexivity that an operator $A : X \rightarrow Y$ fails to be weakly compact if and only if there exists $(x_i) \subset B_X$ so that (Ax_i) dominates (s_i) . Therefore for any operator $A : X \rightarrow Y$ and $K \geq 1$, we define

$$\mathbf{WC}(A, X, Y, K) = \{\emptyset\} \cup \left\{ (x_i)_{i=1}^n \in B_X^{<\mathbb{N}} : (s_i)_{i=1}^n \lesssim_K (Ax_i)_{i=1}^n \right\}.$$

We then let

$$\mathbf{WC}(A, X, Y) = \sup_{K \geq 1} o(\mathbf{WC}(A, X, Y, K)).$$

Then A is weakly compact if and only if $\mathbf{WC}(A, X, Y) < \infty$, which follows immediately from the definition. We make the following easy observations:

Proposition 9.1. (i) *The class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) \leq \omega$ is the ideal of super weakly compact operators.*
(ii) *The class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) < \omega_1$ is an ideal.*

The proof of part (i) is essentially the same as the proof that when (e_i) has property (S') , the class of operators $A : X \rightarrow Y$ so that $\mathbf{NP}_{(e_i)}(A, X, Y) \leq \omega$ is the ideal of all operators all ultrapowers of which fail to preserve a copy of (e_i) .

Of course, part (ii) would be trivial if we restricted our attention to separable domains, since if X is separable and $A : X \rightarrow Y$ is an operator, $\mathbf{WC}(A, X, Y)$ is either countable or ∞ . But since the $\mathbf{NP}_1$ index of an operator cannot be larger than the $\mathbf{WC}$ index, the identity operators on the reflexive examples W_ξ yield weakly compact operators having uncountable $\mathbf{WC}$ index. Thus part (ii) is non-trivial. Our proof of part (ii) follows by another descriptive set theoretic argument. Define the function $f : \mathfrak{L} \rightarrow \mathbf{Tr}$ by

$$(X, Y, \hat{A}) \mapsto \{\emptyset\} \cup \left\{ (k), k^\wedge (n_i)_{i=1}^l : (d_{n_i}(X))_{i=1}^l \in \mathbf{WC}(A, X, Y, k) \right\}.$$

Then by the same methods as in Lemma 8.5, $(X, Y, \hat{A}) \mapsto o(f(X, Y, \hat{A}))$ is a Π_1^1 rank on the Π_1^1 subset $\mathfrak{WC} \cap \mathfrak{L} = f^{-1}(\mathbf{WF})$ such that $o(f(X, Y, \hat{A})) = \mathbf{WC}(A, X, Y) + 1$. With this we establish the analogue of Theorem 8.2. There exists a function $\varphi_{WC} : [1, \omega_1) \times [1, \omega_1) \rightarrow [1, \omega_1)$ so that if $X, Y \in \mathbf{Ban}$ (not necessarily separable), $\xi, \zeta < \omega_1$ are such that $\mathbf{WC}(A, X, Y) \leq \xi$ and $\mathbf{WC}(B, X, Y) \leq \zeta$, then $\mathbf{WC}(A + B, X, Y) \leq \varphi_{WC}(\xi, \zeta)$. The proof is an inessential modification of the proof of Theorem 8.2.

Question 9.2. For which ordinals ξ is the class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) < \xi$

Π_1^1 subset $\mathfrak{WC} \cap \mathfrak{L} = f^{-1}(\mathbf{WF})$ such that $o(f(X, Y, A)) = \mathbf{WC}(A, X, Y) + 1$. With this we establish the analogue of Theorem 8.2. There exists a function $\varphi_{WC} : [1, \omega_1) \times [1, \omega_1) \rightarrow [1, \omega_1)$ so that if $X, Y \in \mathbf{Ban}$ (not necessarily seaparable), $\xi, \zeta < \omega_1$ are such that $\mathbf{WC}(A, X, Y) \leq \xi$ and $\mathbf{WC}(B, X, Y) \leq \zeta$, then $\mathbf{WC}(A + B, X, Y) \leq \varphi_{WC}(\xi, \zeta)$. The proof is an inessential modification of the proof of Theorem 8.2.

Question 9.3. *For which ordinals ξ is the class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) \leq \xi$ an ideal?*

9.3. Other applications of $\mathfrak{L}$ and $\mathfrak{L}_1$. In some cases, it is perhaps more convenient to code only the operators having norm not exceeding 1. One convenience of $\mathfrak{L}_1$ is the following concerning the Szlenk index $Sz(A)$ of an operator. Recall that if X is a separable Banach space, $A : X \rightarrow Y$ is an Asplund operator if and only if $A^*B_{Y^*}$ is norm separable. Let $\mathfrak{A}$ denote the ideal of Asplund operators.

2 Definitions

Let $s_i = e_1 + \cdots + e_i$ be the summing basis of c_0 . We use the tail-sum convention

$$\left\| \sum_{i=1}^n a_i s_i \right\| = \max_{1 \leq j \leq n} \left| \sum_{i=j}^n a_i \right|.$$

For an operator $A : X \rightarrow Y$ and $K > 1$, let $\mathbf{WC}(A, K)$ be the tree of finite sequences $(x_i)_{i=1}^n \subset B_X$ such that

$$\left\| \sum_i a_i s_i \right\| \leq K \left\| \sum_i a_i A x_i \right\| \quad (a_i \in \mathbb{R}).$$

The index is

$$\mathbf{WC}(A) = \sup_{K > 1} o(\mathbf{WC}(A, K)),$$

with order ∞ for ill-founded trees.

3 Finite Levels

Theorem 3.1 (finite ranks). *For every bounded operator $A : X \rightarrow Y$ and every integer $n \geq 1$,*

$$\mathbf{WC}(A) \leq n \iff \text{rank}(A) \leq n - 1.$$

Equivalently, if $\text{rank}(A) = r < \infty$, then $\mathbf{WC}(A) = r + 1$, while every infinite-rank operator satisfies $\mathbf{WC}(A) \geq \omega$.

Proof. If $\text{rank}(A) = r$ and $(x_i)_{i=1}^m \in \mathbf{WC}(A, K)$, then $(Ax_i)_{i=1}^m$ dominates the linearly independent sequence $(s_i)_{i=1}^m$. Hence $m \leq r$, so every node has length at most r and the tree order is at most $r + 1$.

Conversely, choose r linearly independent vectors $y_i = Ax_i$ with $x_i \in B_X$. Any two bases of an r -dimensional space are equivalent, so for some K the sequence $(y_i)_{i=1}^r$ dominates $(s_i)_{i=1}^r$. This gives a branch of length r , hence order at least $r + 1$. If A has infinite rank, repeat this for every finite length and take the supremum over K . $\square$

Corollary 3.2. *No finite cutoff $n \geq 2$ is closed under sums. The cutoff $n = 1$ is only the zero-operator class and is not an operator ideal under the convention that ideals contain all finite-rank operators.*

4 Exact Triangular Arrays

Proposition 4.1 (triangular criterion). *For vectors $y_1, \dots, y_n \in Y$ and $K \geq 1$, the following are equivalent.*

- (i) $(s_i)_{i=1}^n \lesssim_K (y_i)_{i=1}^n$.
- (ii) *There are $f_1, \dots, f_n \in KB_{Y^*}$ such that*

$$f_j(y_i) = \begin{cases} 0, & i < j, \\ 1, & i \geq j. \end{cases}$$

Proof. If (i) holds, the map $Ty_i = s_i$ has norm at most K . Extend e_j^*T by Hahn–Banach. Conversely, choose j where the tail-sum norm of $\sum_i a_i s_i$ is attained; applying f_j gives the lower estimate. $\square$

Lemma 4.2 (summable triangular errors). *Let $y_1, \dots, y_n \in Y$. Suppose $f_1, \dots, f_n \in Y^*$, $\|f_j\| \leq L$, and*

$$|f_j(y_i) - \mathbf{1}_{\{i \geq j\}}| \leq \varepsilon_i \quad (1 \leq i, j \leq n).$$

If $E = \sum_i \varepsilon_i < 1/2$, then

$$(s_i)_{i=1}^n \lesssim_{L/(1-2E)} (y_i)_{i=1}^n.$$

Proof. For scalars a_i , put $T_j = \sum_{i=j}^n a_i$ and $M = \max_j |T_j|$. Then $a_i = T_i - T_{i+1}$, so $|a_i| \leq 2M$. If $|T_{j_0}| = M$, then

$$\left| f_{j_0} \left(\sum_i a_i y_i \right) - T_{j_0} \right| \leq \sum_i |a_i| \varepsilon_i \leq 2ME.$$

Thus $L \|\sum_i a_i y_i\| \geq (1 - 2E)M$. $\square$

Proposition 4.3 (fixed mesh is insufficient). *For every $\delta > 0$ there are arbitrarily large n , vectors $y_1, \dots, y_n \in \ell_\infty^n$, and coordinate functionals $f_1, \dots, f_n$ such that*

$$|f_j(y_i) - \mathbf{1}_{\{i \geq j\}}| < \delta$$

for all i, j , while (y_i) is linearly dependent and hence cannot dominate the summing basis.

Proof. Let $H_{ji} = \mathbf{1}_{\{i \geq j\}}$. Choose alternating tail sums $T_j = (-1)^j$, $T_{n+1} = 0$, and set $a_i = T_i - T_{i+1}$. Then $\sum_i |a_i| = 2n - 1 =: L_0$ and $\sum_i a_i H_{ji} = T_j$. Put

$$E_{ji} = -T_j \operatorname{sgn}(a_i)/L_0, \quad y_i = (H_{ji} + E_{ji})_{j=1}^n.$$

For large n , $|E_{ji}| < \delta$, but $\sum_i a_i y_i = 0$. Coordinate functionals give the required approximate triangular matrix. $\square$

5 Structural Tools

Proposition 5.1 (composition). *If $U : Y \rightarrow Z$, $A : X \rightarrow Y$, and $V : W \rightarrow X$ are bounded, then*

$$\text{WC}(UAV) \leq \text{WC}(A).$$

Thus for every $\xi \geq \omega$, the class $\mathcal{W}_\xi$ contains finite-rank operators and is stable under scalar multiplication and two-sided composition. Addition is the only unresolved ideal axiom at fixed infinite levels.

Proof. A node for UAV maps to a rescaled node for A , with only the domination constant changed. The finite-rank assertion follows from Theorem 3.1. $\square$

Proposition 5.2 (direct-sum lower bound). *Let $A : X \rightarrow Y$ and $B : W \rightarrow Z$, and use*

$$A \oplus B : X \oplus_\infty W \rightarrow Y \oplus_1 Z.$$

If the nonempty parts of $\text{WC}(A, K)$ and $\text{WC}(B, K)$ contain B -subtrees of orders α and β , then the nonempty part of $\text{WC}(A \oplus B, K)$ contains a subtree of order $\beta + \alpha$.

Proof. Concatenate an A -node followed by a B -node. The summing norm of a concatenated scalar vector is at most the sum of the two separate summing norms, while the $Y \oplus_1 Z$ norm gives the corresponding sum of lower estimates. The standard leaf-replacement derivative calculation gives order $\beta + \alpha$. $\square$

Theorem 5.3 (successor obstruction). *Let $\xi \geq 2$ be a successor ordinal. If there is an operator A with $\text{WC}(A) = \xi$, then $\mathcal{W}_\xi$ is not closed under addition.*

Proof. Write $\xi = \eta + 1$. For some K , the nonempty part of $\text{WC}(A, K)$ has a subtree of order η . Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be the identity, whose one-node B -tree has order 1. Proposition 5.2 gives $\text{WC}(F \oplus A) \geq \xi + 1$, while

$$F \oplus A = (F \oplus 0) + (0 \oplus A)$$

and both summands have index at most ξ . $\square$

6 Comparison With Causey's James Index

Let $\mathcal{J}(A, \varepsilon)$ be the tree of finite sequences $(x_i) \subset B_X$ whose images are ε -convexly separated at every split. Causey's James index is

$$J(A) = \sup_{\varepsilon > 0} o(\mathcal{J}(A, \varepsilon)).$$

Proposition 6.1. *For every operator A , $\text{WC}(A) \leq J(A)$.*

Proof. If $(x_i) \in \text{WC}(A, K)$, then for every split m , the difference of a convex combination of $\{Ax_i : i \leq m\}$ and one of $\{Ax_i : i > m\}$ has coefficient vector with summing norm at least 1. Hence the two convex hulls are $1/K$ -separated. $\square$

Remark 6.2. Causey proved that the classes

$$\{A : J(A) \leq \omega^{\omega^\eta}\}$$

are closed operator ideals [2]. Proposition 6.1 explains why this is relevant to Question 9.3, but it does not solve it: finite-rank nonzero operators have $J(A) = \omega$, while $\text{WC}(A) = \text{rank}(A) + 1$.

7 Finite Localization and Fixed-Mesh Ramsey Input

The analytic obstruction is not finite-dimensional. Rosenthal's wide-(s) theory gives the following localization in finite length.

Proposition 7.1 (finite localization). *For every $R < \infty$, $\varepsilon > 0$, and $k \in \mathbb{N}$, there are $N \in \mathbb{N}$ and $L < \infty$ such that every ε -convexly separated sequence $(y_i)_{i=1}^N \subset RB_Y$ in any Banach space has an L -wide-(s) subsequence of length k , hence a subsequence dominating the summing basis with constant L .*

Proof. If not, take longer and longer counterexamples and form an ultraproduct. The resulting infinite convexly separated sequence has no weakly convergent subsequence; otherwise Mazur convex blocks would contradict convex separation. Rosenthal's theorem yields a wide-(s) subsequence. The finite inequalities defining its first k terms transfer back to large finite coordinates. $\square$

Causey–Doebele identify the ordinals with full-rank finite-color homogeneity for comparable pairs in well-founded trees as exactly

$$\omega^{\omega^\eta}.$$

Together with Causey's extended-pair stabilization, this gives fixed-mesh control of the scalar arrays arising from a triangular representation of $(A + B)x_t$. Proposition 4.3 shows why this fixed mesh is not yet enough: the summing-basis lower estimate needs errors that are summable along a branch.

8 Conditional Dependency: Summable Fusion

Let M_λ be the standard B -tree of order λ .

Conditional Dependency 8.1 (summable fusion $\text{SF}(\lambda)$). An infinite ordinal λ has property $\text{SF}(\lambda)$ if the following holds. Given $R < \infty$, a bounded array

$$u = (u_A, u_B) : \{(p, q, v) : p \neq q, p, q \preceq v, v \in \text{MAX}(M_\lambda)\} \rightarrow [-R, R]^2,$$

and a positive summable sequence (η_n) , there are $\alpha, \beta \in [-R, R]^2$, monotone maps

$$r, c, a : M_\lambda \rightarrow M_\lambda, \quad e : \text{MAX}(M_\lambda) \rightarrow \text{MAX}(M_\lambda),$$

such that along every branch

$$r(t_1) \prec c(t_1) \prec a(t_1) \prec r(t_2) \prec c(t_2) \prec a(t_2) \prec \cdots, \quad a(v) \preceq e(v),$$

and, for all $s, t \preceq v$,

$$s \prec t \Rightarrow \|u(c(s), r(t), e(v)) - \alpha\|_\infty < \eta_{|s|},$$

$$t \preceq s \Rightarrow \|u(c(s), r(t), e(v)) - \beta\|_\infty < \eta_{|s|},$$

$$\|u(c(s), a(v), e(v)) - \alpha\|_\infty < \eta_{|s|}.$$

9 Conditional Ideal Theorem

Theorem 9.1. *If an infinite ordinal λ satisfies $\text{SF}(\lambda)$, then*

$$\mathcal{W}_\lambda = \{A : \text{WC}(A) \leq \lambda\}$$

is an operator ideal.

Proof. By Proposition 5.1, only addition must be proved. We argue contrapositively. Let $C = A + B$ and suppose $\text{WC}(C) > \lambda$. Pick $K > 1$ and a branch-preserving family $(x_t)_{t \in M_\lambda} \subset B_X$ whose every branch lies in $\text{WC}(C, K)$. By Proposition 4.1, for every maximal v and every $q \preceq v$, there is $\phi_{q,v} \in KB_{Y^*}$ with

$$\phi_{q,v}(Cx_p) = \begin{cases} 0, & p \prec q, \\ 1, & q \preceq p \preceq v. \end{cases}$$

Define

$$u_D(p, q, v) = \phi_{q,v}(Dx_p), \quad D \in \{A, B\}.$$

Choose $\eta_n > 0$ with $\sum_n \eta_n \leq 1/16$, and apply $\text{SF}(\lambda)$. Since $\eta_n \rightarrow 0$ and M_λ has arbitrarily long branches, the exact equations for $C = A + B$ force

$$\alpha_A + \alpha_B = 0, \quad \beta_A + \beta_B = 1.$$

Thus for one $D \in \{A, B\}$,

$$d := \beta_D - \alpha_D, \quad |d| \geq 1/2.$$

For $t \preceq v$, set

$$g_{t,v} = \frac{\phi_{r(t),e(v)} - \phi_{a(v),e(v)}}{d}.$$

Then $\|g_{t,v}\| \leq 4K$. The summable-fusion inequalities give an approximately triangular array for $Dx_{c(s)}$ with column errors $4\eta_{|s|}$. Since $\sum_s 4\eta_{|s|} \leq 1/4$ along each selected branch, Lemma 4.2 implies that every selected branch $8K$ -dominates the summing basis. Hence c embeds M_λ branchwise into $\text{WC}(D, 8K)$, so $\text{WC}(D) > \lambda$. Therefore if both $\text{WC}(A)$ and $\text{WC}(B)$ are at most λ , then $\text{WC}(A + B) \leq \lambda$. $\square$

10 Alternative Completion Route

The previous packet's James-to-wide-(s) bridge remains a viable alternate target:

$$J(A) > \omega\alpha \implies \text{WC}(A) > \alpha.$$

If this holds, then at every $\lambda = \omega^{\omega^\eta}$ with $\eta > 0$, the implication $\text{WC}(A) \leq \lambda \implies J(A) \leq \lambda$ follows because $\omega\lambda = \lambda$. Causey's James-index ideal theorem and Proposition 6.1 would then give the same positive conclusion for $\mathcal{W}_\lambda$. The bottom level ω is already known directly from the source paper.

11 Present Status

The current evidence suggests the candidate pattern

$$\lambda = \omega^{\omega^\eta}$$

for the infinite ideal cutoffs, but this is not proved here. What is proved is:

- (1) all finite levels are determined exactly by rank;
- (2) ω is positive by the source paper;
- (3) successor levels fail whenever the corresponding exact WC value occurs;

- (4) fixed-mesh pair stabilization is available in the literature at the natural ordinals;
- (5) fixed mesh is provably insufficient;
- (6) the ideal theorem follows from the explicit summable-fusion property $\text{SF}(\lambda)$.

The supplied report records searches by the source notation, source title, “summing basis”, “wide- (s) ”, “James index”, and pair-Ramsey terminology, and did not locate a published full answer to Question 9.3. This upgrade did not perform a fresh external web search.

12 Human Review Recommendation

The first review target is $\text{SF}(\omega^{\omega^n})$. A proof would need to strengthen known finite-color comparable-pair stabilization from fixed mesh to a prescribed summable mesh without losing tree rank. A counterexample to this fusion principle would indicate where the natural classification pattern can fail.

References

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